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Education

Universitas Gunadarma

B.Sc. in Informatics, GPA 3.62 / 4.00

2022 - present

Depok, West Java

Undergraduate Thesis

Home Power Usage Management System with a Multimodal AI Agent (Groq)

In Progress

- Built an MQTT pipeline where ESP32 nodes with an LD2410C presence radar and a PZEM-004T meter publish room and power data, streams updates to a Vue 3 dashboard over WebSocket STOMP, and stores power history in InfluxDB.
- Added a Groq-powered energy advisor that streams answers over Server-Sent Events and handles image queries, Whisper voice input, and PDF, DOCX and TXT questions in per-user conversation threads with background long-term memory.
- Secured the API with JWT access and rotating refresh tokens, Redis sliding-window rate limits, and a 10-minute logout after 5 failed logins, then closed the OWASP API and IoT checklists until a ZAP scan of 112 endpoints reported no high or medium findings.
- Covered the backend with 12 unit tests, 9 Selenium UI tests and 11 Cucumber scenarios, gated a k6 load test at p95 under 500 ms, added 30 Vitest frontend tests, and provisioned Grafana dashboards and Loki log shipping as code.

Projects

AI Agent Chat Assistant

Jun 2026 - Jul 2026

- Wrote a fullstack chat app where the agent picks a search or page reading tool on its own, streams the answer to the browser over Server-Sent Events, and shows each reasoning step in the thread.
- Accepted image, voice and document messages with Groq vision, Whisper for audio, Apache Tika for PDF and Word text, caching repeated answers and retrying on rate limits.
- Signed users in with Google OAuth2 or a Resend magic link, kept sessions alive with a short-lived JWT and a rotating refresh token, and capped the AI endpoints at 30 requests a minute per IP.

PT. Pandawa Digital Mandiri Website

Nov 2025 - May 2026

- Built the admin console and the public site of a client and service request platform on Angular 19, Spring Boot 3 and PostgreSQL, with JWT logins split between super admin and manager roles.
- Scored incoming leads with Gemini AI from the request budget, timeline and message, cached each analysis in Redis for an hour, and held the endpoint at 15 calls per admin every 2 minutes.
- Routed notifications through RabbitMQ, so a verified or rejected request reaches admins over STOMP WebSocket and the client as a Resend email.

Car Rental Management System

April 2024

- Developed a desktop application using Java and Swing for managing car rental operations, including car availability checks, customer data management, and transaction recording.
- Consolidated multiple rental management functions (car data, customer data, rental status) into a single application.

Technical Skills

Tech Stack: Java, Spring Boot, Vue.js, Angular, TypeScript, PostgreSQL, Redis, Docker

IoT and embedded: ESP32, MQTT with Mosquitto, LD2410C mmWave radar, PZEM-004T, InfluxDB, relay based actuation